Supplementary Material

This document contains all appendices referenced in the main manuscript, including detailed simulation settings, results for unprofitable planning, and sensitivity analyses of key capacity assumptions. All tables, figures, and explanatory text referenced in the main manuscript are included for completeness.
The remainder of the Supplementary Material is organised as follows. Section Appendix A presents the simulation settings and numerical implementation details. Section Appendix B shows the results for unprofitable planning. Section Appendix C reports the sensitivity analysis of capacity settings.

Appendix A. Simulation settings

This appendix summarises the simulation settings and numerical implementation details of the modelling framework. It documents the iterative solution procedure between the long-term investment model and the short-term operational model, including convergence criteria, iterations, coalitions of Shapley values, computational time, and the simulation environment used for implementation.

Appendix A.1. Competitive planning implementation (Nash–Cournot)

The game-theory model (GTM) is solved iteratively in coordination with the short-term Optimal Power and Gas Flow (OPGF) model. The solution algorithm, illustrated in Figure 1 (a), follows these steps:

1. Initialise system parameters and decision variables.
2. Run the GTM to determine optimal investment and generation decisions endogenously, where the upper-layer long-term planning is formulated as a Mathematical Program with Equilibrium Constraints (MPEC), in which strategic interactions are represented through complementarity conditions, and solved using the PATH solver [1].
3. Use GTM outputs as input to the OPGF model to check short-term operational feasibility under technical network constraints. The electrical network is simulated and optimised using Pandapower [2], while the hydrogen/gas network is simulated using Pandapipes [3], which provides functionalities to model gas mixtures and steady-state operation.

4. Check convergence: the process iterates until the difference between successive energy mixes from OPGF and GTM falls below a predefined tolerance, $\varepsilon = 0.01$, i.e., $|Q^{(i+1)}_{\text{OPGF}} - Q^{(i)}_{\text{GTM}}| < \varepsilon$.
5. The maximum number of iterations is set to 10, consistent with the default limit used in the Newton-based power and gas flow solvers; if convergence is not reached within this limit, the last feasible solution is used for reporting.
6. Upon convergence, final metrics including supply, demand, costs, and emissions are computed, enabling assessment of system-wide efficiency for the analysed heating decarbonisation pathways.

Appendix A.2. Cooperative planning implementation using Shapley values

The cooperative approach determines energy allocations among players based on Shapley values, reflecting each player's contribution to joint welfare. While the allocations are computed in a decentralised manner, they operationally resemble a centralised allocation, as players form coalitions to maximise joint welfare and share demand according to well-defined contribution-based weights. Unlike the competitive GTM, there are no strategic interactions or complementarity constraints, and the process avoids conflicting incentives that could cause oscillations. The flowchart of the solution is illustrated in Figure 1 (b), and the solution algorithm is as follows:

1. Initialise system parameters, player capacities, and decision variables.
2. Run the cooperative allocation to determine investment and generation decisions endogenously, where energy shares are computed via Shapley values to maximise joint welfare.
3. Use the Shapley-based allocations as input to the OPGF model to verify operational feasibility under technical network constraints. The electrical network is simulated with Pandapower, and the hydrogen/gas network with Pandapipes.
4. Check convergence: the process iterates until successive energy allocations are consistent with the OPGF solution within a tolerance of $\varepsilon = 0.01$.
5. The maximum number of iterations is set to 10, consistent with the default limit used in the Newton-based power and gas flow solvers; if convergence is not reached within this limit, the last feasible solution is used for reporting.
6. Upon convergence, final metrics including supply, demand, costs, and emissions are computed, enabling assessment of system-wide efficiency

for the analysed heating decarbonisation pathways. The framework remains cooperative and non-strategic, as allocations are determined by Shapley contributions rather than individual profit-maximising behaviour.

Appendix A.3. Benchmark planning implementation using a centralised optimised approach

The centralised optimised benchmark is implemented as a single, centralised optimisation problem that jointly determines long-term investment and short-term operational dispatch. Unlike the GTM, the benchmark assumes a single planner and does not include strategic interactions or complementarity conditions between market actors. Key aspects of the benchmark implementation are:

1. Comparability: The benchmark uses the same datasets as the GTM, including demand time series, generation types, installed capacities, and network parameters, ensuring a fair comparison of outcomes.
2. Solver and formulation: The benchmark is implemented in Pyomo, a Python-based optimisation modelling environment, and formulated as a standard non-linear programming (NLP) problem. The resulting optimisation is solved using IPOPT (Interior Point OPTimizer). All operational constraints, including generation limits, network capacities, and power balance, are enforced directly within the same centralised optimisation problem.
3. Convergence and tolerance criteria: To ensure consistency with the competitive and cooperative approaches, the benchmark optimisation iterates between the long-term planning and short-term operational layers. The maximum number of iterations is set to 10, and the tolerance is set to $\epsilon = 0.01$. If convergence is not achieved within this limit, the last feasible solution is retained for reporting.
4. Outputs and metrics: For each scenario, the benchmark provides system-level metrics such as total system cost, net present value (NPV), generation mix, operational dispatch, and CO_2 emissions, which are reported in Table 3 for comparison with the competitive and cooperative approaches.

This implementation ensures that differences observed between the GTM and the benchmark reflect solely the presence or absence of strategic interactions, while keeping all other modelling assumptions consistent.

Appendix A.4. Computational performance

Table A.1 reports the computational time required to solve a single simulation snapshot for each planning approach and investment scenario. The reported values correspond to the total wall-clock time needed to reach convergence of the long-term planning model in coordination with the short-term OPGF model, including all iterative updates where applicable. All simulations were executed on the same computational platform to ensure comparability across approaches. The simulations were performed on a workstation equipped with an AMD Ryzen AI 7 350 processor with Radeon 860M, 8 cores (16 threads) running at 2.0 GHz, and 24 GB of RAM. The computational times for all planning approaches across scenarios had no significant impact on our comparative analysis.

Table A.1: Computational time (seconds) required for each planning approach and investment scenario, evaluated with the same simulation snapshot, (peak demand hour).

Planning approach	Uniform +25GW/Tech.	High RES– High H_2	High RES– Low H_2	High H_2– Low BESS
Competitive	14.64	10.65	27.11	12.39
Cooperative	10.33	10.28	12.91	12.99
Benchmark	10.55	11.49	13.09	12.88

Appendix B. Results for unprofitable planning

This appendix presents the results of unprofitable planning scenarios for Competitive, Cooperative, and Benchmark approaches over a 25-year investment span, using the same assumptions as in the profitable planning, whose results are shown in Table 3. These scenarios correspond to conditions under which the expected Net Present Value (NPV) of the system is negative, resulting from high capital expenditures, adverse commodity price assumptions (electricity, hydrogen, natural gas, and CO_2), and suboptimal technology mixes. The unprofitable conditions were inferred from Figure 8 (a–c), which show the sensitivity of NPV to fluctuations in electricity, hydrogen, and natural gas prices and allow identification of the price levels where NPV becomes negative. Table B.1 then summarises the resulting system performance under these unprofitable conditions for each planning paradigm.

Complementing the quantitative results, Figures B.1–B.3 illustrate the corresponding energy generation mixes under unprofitable planning for the Competitive, Cooperative, and Benchmark approaches, providing additional insight into how adverse economic conditions influence technology deployment and system structure across scenarios.

Table B.1 summarises the unprofitable planning scenarios, highlighting substantial differences in economic outcomes across the three planning approaches. Competitive planning exhibits consistently large negative NPVs across all scenarios, often accompanied by relatively high operational costs and, in some cases, significant residual CO_2 emissions, reflecting elevated financial risk under decentralised decision-making. Cooperative planning substantially reduces these losses through coordinated investment and operation, resulting in the smallest magnitude of negative NPVs, lower operational costs, and generally limited CO_2 emissions. This indicates that cooperative strategies are more effective at mitigating downside economic risk under adverse investment conditions. Benchmark planning yields intermediate outcomes between the competitive and cooperative approaches. While the benchmark generally achieves lower operational costs and reduced emissions compared to competitive planning, it still experiences sizable negative NPVs in most scenarios. This suggests that although centralised planning improves operational efficiency under unprofitable investment conditions, it does not fully eliminate financial losses and remains less robust than cooperative planning in absorbing economic uncertainty.

Under unprofitable planning conditions, the cooperative approach consistently exhibits the smallest negative Net Present Value across all scenarios, indicating greater economic robustness relative to competitive and benchmark planning. As a result, cooperative planning requires the lowest level of policy support to sustain participation, as evidenced by Table 5 and the detailed results reported in this appendix. This demonstrates that coordinated investment and operation not only reduce system costs and emissions, but also minimise public intervention requirements under adverse market conditions.

Appendix C. Results for sensitivity analysis of the capacity settings

To address the influence of capacity limits (see Table 2) on the planning results (as reported in Table 3), we conducted targeted sensitivity analysis by

Table B.1: Comparison of simulation results for unprofitable planning under Competitive, Cooperative, and Benchmark approaches (Investment span: 25 years). Electric peak demand is 102.17 GWh, and hydrogen peak demand is 14.27 GWh. The Benchmark planning follows a centralised optimised approach.

Competitive planning	Investment scenarios 2025–2050			
	Uniform +25GW/Tech.	High RES– High H_2	High RES– Low H_2	High H_2– Low BESS
Total demand [GWh]	182.75	152.06	117.75	187.67
Electric demand [GWh]	149.44	130.17	103.10	158.17
Hydrogen demand [GWh]	33.31	21.88	14.65	29.50
Total energy supply [GWh]	182.75	152.06	117.75	187.67
Total operational cost [m£]	2.03	1.70	1.15	3.49
Total CO_2 emissions [tonnes]	0	0	0	10785
Net Present Value (NPV) [bn£]	-334	-184	-120	-293
Cooperative planning	Investment scenarios 2025–2050			
	Uniform +25GW/Tech.	High RES– High H_2	High RES– Low H_2	High H_2– Low BESS
Total demand [GWh]	145.63	140.51	117.15	145.22
Electric demand [GWh]	118.64	118.63	102.50	118.63
Hydrogen demand [GWh]	26.99	21.88	14.65	26.59
Total energy supply [GWh]	145.63	140.51	117.15	145.22
Total operational cost [m£]	1.19	1.49	0.90	1.26
Total CO_2 emissions [tonnes]	0	1230	0	0
Net Present Value (NPV) [bn£]	-69	-133	-86	-118
Central planning (Benchmark)	Investment scenarios 2025–2050			
	Uniform +25GW/Tech.	High RES– High H_2	High RES– Low H_2	High H_2– Low BESS
Total demand [GWh]	182.75	152.06	117.75	187.67
Electric demand [GWh]	149.44	130.17	103.10	158.17
Hydrogen demand [GWh]	33.31	21.88	14.65	29.50
Total energy supply [GWh]	182.75	152.06	117.75	187.67
Total operational cost [m£]	1.86	1.61	1.05	2.60
Total CO_2 emissions [tonnes]	0	0	0	1516
Net Present Value (NPV) [bn£]	-301	-171	-107	-363

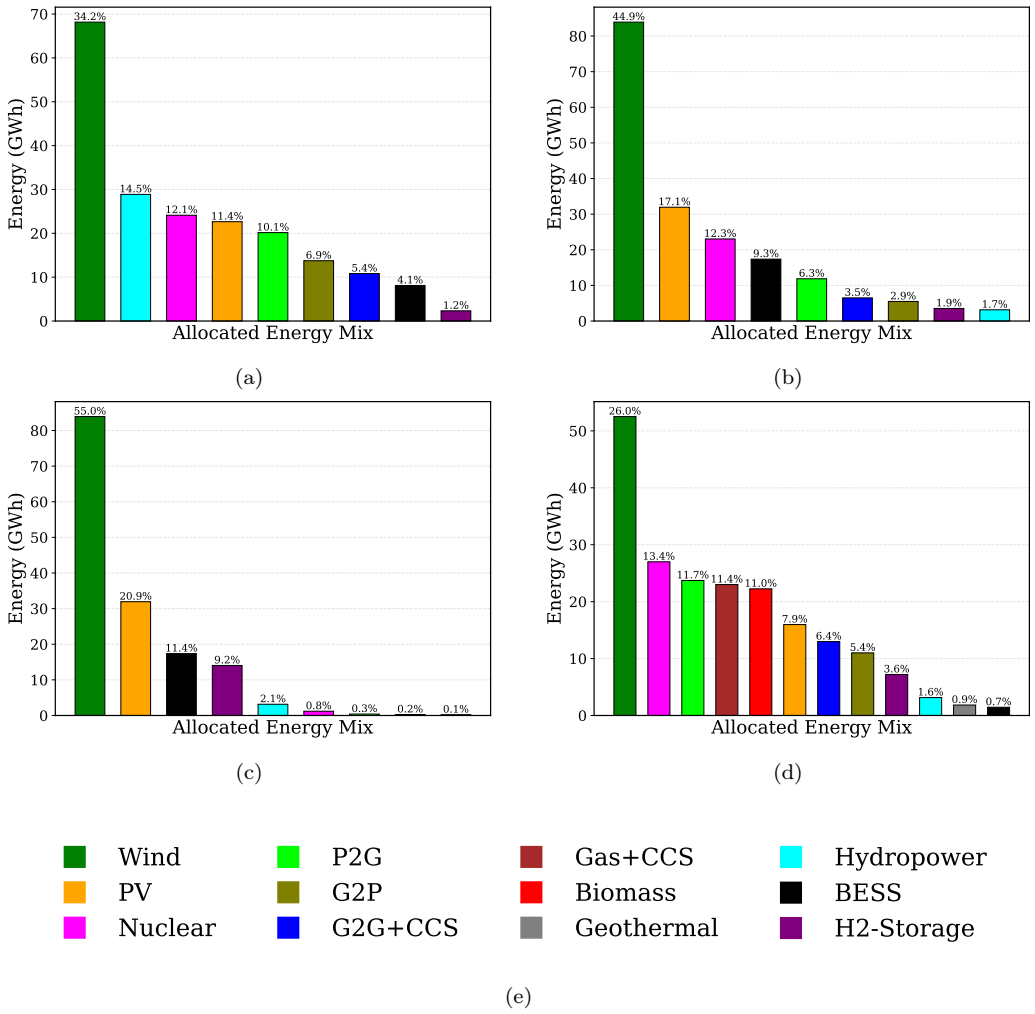


Figure B.1: Competitive planning of energy generation mix (GWh) for 2050 peak demand under unprofitable planning: (a) Uniform +25 GW, (b) High RES–High H_2, (c) High RES–Low H_2, (d) High H_2–Low BESS. (e) Legend for Technology Types.

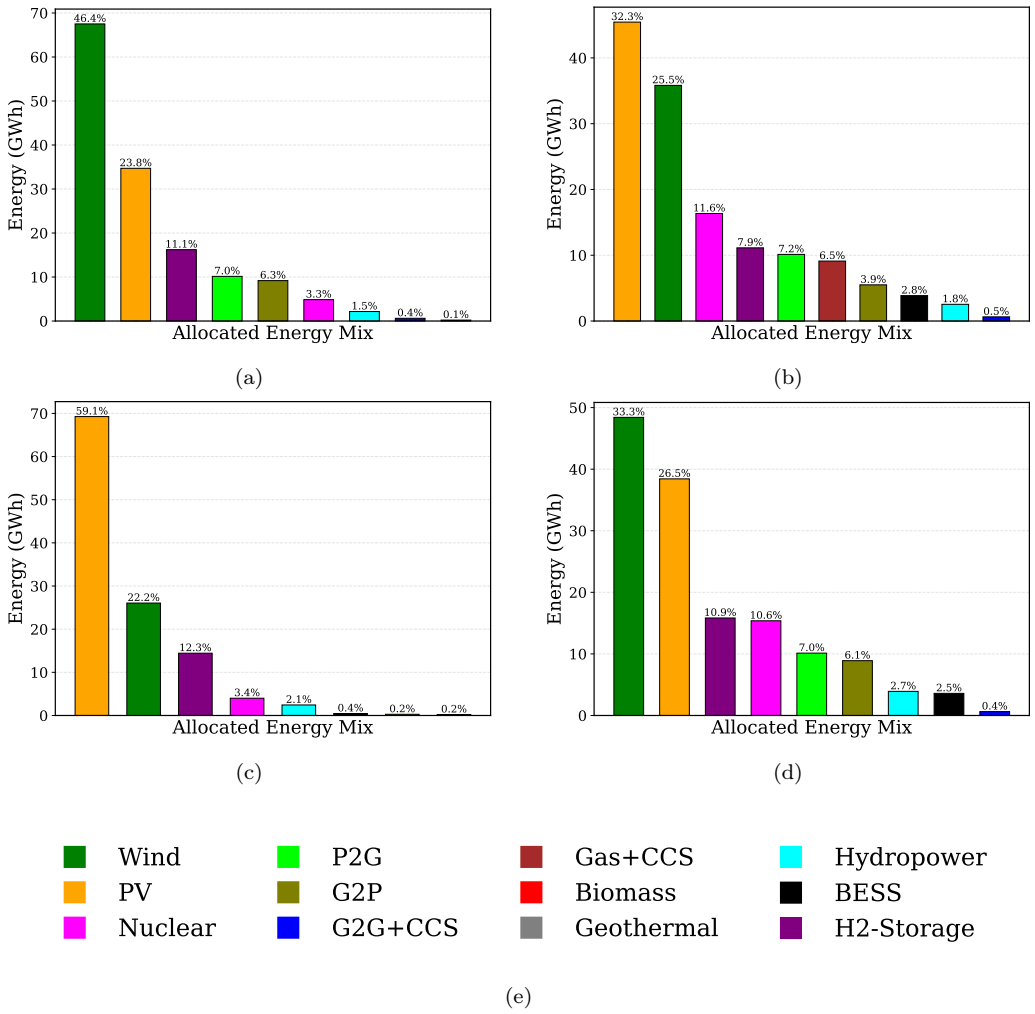
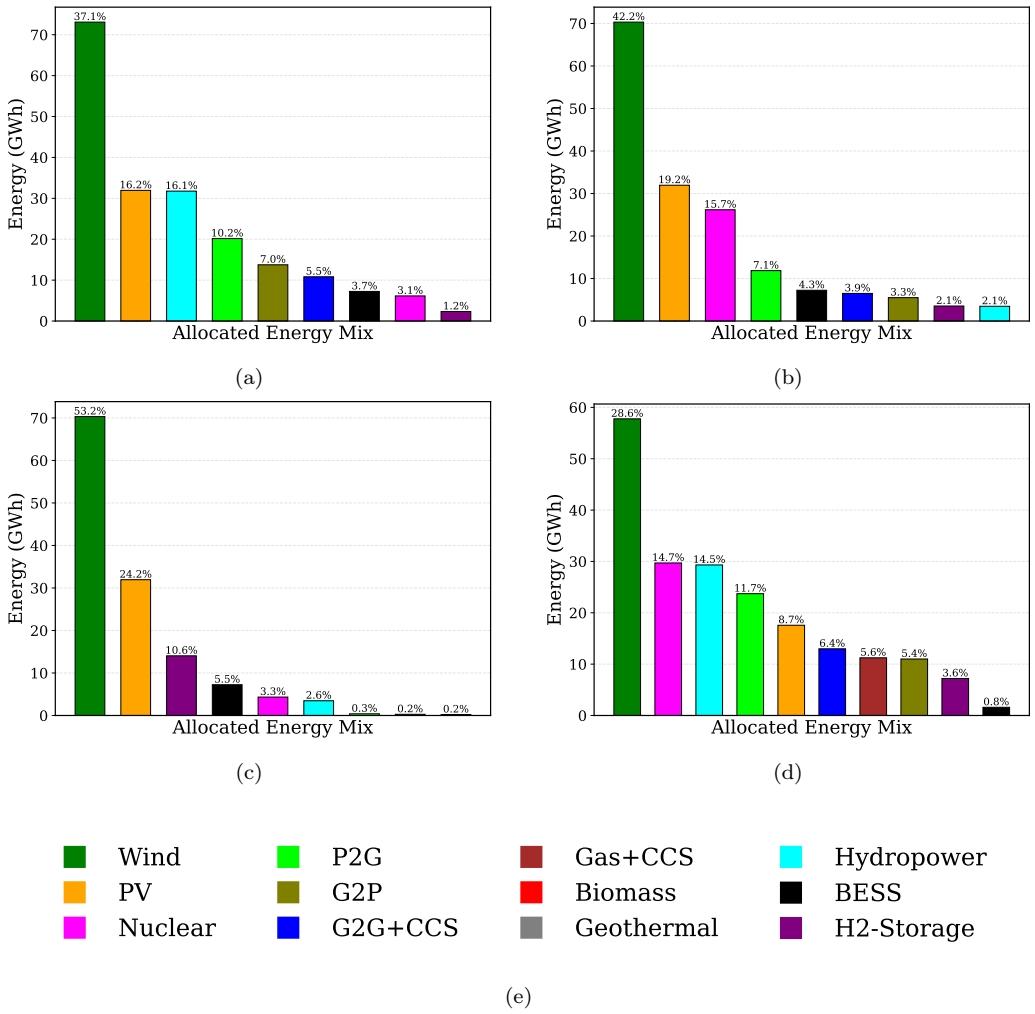


Figure B.2: Cooperative planning of energy generation mix (GWh) for 2050 peak demand under unprofitable planning: (a) Uniform +25 GW, (b) High RES–High H_2, (c) High RES–Low H_2, (d) High H_2–Low BESS. (e) Legend for Technology Types.



(e)

Figure B.3: Benchmark planning of energy generation mix (GWh) for 2050 peak demand under unprofitable planning: (a) Uniform +25 GW, (b) High RES–High H_2, (c) High RES–Low H_2, (d) High H_2–Low BESS. (e) Legend for Technology Types.

varying the baseline capacity limits of key technologies (renewables, nuclear, hydrogen, energy storage) by ±20%. The results in Table C.1 demonstrate the robustness of our main findings across competitive, cooperative, and benchmark approaches. Increasing capacity limits (+20%) generally maintains the optimal baseline solution, while tightening limits (−20%) occasionally increases operational costs but does not materially alter key system performance metrics or lead to infeasibility in some cases.

Table C.1: Sensitivity of planning results to capacity settings across competitive, cooperative, and benchmark approaches, with two scenarios of hydrogen integration, High: (High RES–High H_2) and Low: (High RES–Low H_2) at the. Electric peak demand is 102.17 GWh, and hydrogen peak demand is 14.27 GWh. The Benchmark planning follows a centralised optimised approach.

Metric	−20%		**100% (baseline)**		+20%	
	High H_2	Low H_2	High H_2	Low H_2	High H_2	Low H_2
Competitive						
Total demand [GWh]	152.06	117.75	**152.05**	**117.75**	152.05	117.75
Electric demand [GWh]	130.17	103.10	**130.17**	**103.10**	130.17	103.10
Hydrogen demand [GWh]	21.88	14.65	**21.88**	**14.65**	21.88	14.65
Total supply [GWh]	152.06	117.75	**152.05**	**117.75**	152.05	117.75
Total operational cost [m£]	1.88	1.33	**2.09**	**1.13**	1.91	1.13
Total CO_2 emissions [tonnes]	0	0	**0**	**0**	0	0
Net Present Value (NPV) [bn£]	706	622	**975**	**689**	1044	699
Cooperative						
Total demand [GWh]	132.94	116.81	**132.95**	**116.99**	132.95	117.18
Electric demand [GWh]	114.73	102.31	**114.73**	**102.50**	114.73	102.68
Hydrogen demand [GWh]	18.21	14.50	**18.23**	**14.50**	18.23	14.50
Total supply [GWh]	132.94	116.81	**132.95**	**116.99**	132.95	117.18
Total operational cost [m£]	1.34	1.08	**1.41**	**1.14**	1.40	1.13
Total CO_2 emissions [tonnes]	0	0	**0**	**0**	0	0
Net Present Value (NPV) [bn£]	510	465	**485**	**442**	485	446
Benchmark						
Total demand [GWh]	152.06	117.75	**152.06**	**117.75**	152.06	117.75
Electric demand [GWh]	130.17	103.10	**130.17**	**103.10**	130.17	103.10
Hydrogen demand [GWh]	21.88	14.65	**21.88**	**14.65**	21.88	14.65
Total supply [GWh]	152.06	117.75	**152.06**	**117.75**	152.06	117.75
Total operational cost [m£]	1.78	1.22	**1.56**	**1.13**	1.49	1.13
Total CO_2 emissions [tonnes]	0	0	**0**	**0**	0	0
Net Present Value (NPV) [bn£]	745	661	**824**	**700**	856	699

References

[1] S. P. Dirkse, M. C. Ferris, The path solver: a nommonotone stabilization scheme for mixed complementarity problems, Optimization methods and software 5 (2) (1995) 123–156.

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[3] D. Lohmeier, D. Cronbach, S. R. Drauz, M. Braun, T. M. Kneiske, Pandapipes: An open-source piping grid calculation package for multi-energy grid simulations, Sustainability 12 (23) (2020) 9899.